

Ecommerce Purchases Exercise Overview

Hi Guys,

After a crash course on pandas for data analysis, its time to do some practice!!!

Because of privacy issues, I have created a fake dataset here with 30K entries.

The situation is, customers are providing some personal information while purchasing stuff on-line or in-store. For some reasons, your client wants to know the answer to some of his questions from the dataset, let's try to help him!.

Feel free to consult the solutions if needed. Please note, the tasks given in the exercises, can be solved in different ways. Try your best answer and compare with the solutions.

👉 The dataset is provided in the course downloads.

1. Import Pandas and Read the csv file.

```
In [2]: import pandas as pd
df=pd.read_csv("Cust_Purch_FakeData.csv")
```

2. Its good idea to see how the data look like, display first 5 rows of your data-set.

```
In [3]: # Code here please
df.head()
```

Out[3]:

	prefix	first	last	email	gender	age	company	profession	phone	postal	province	cc_no	cc_exp	cc_t
0	Dr.	Ray	Morton	sebvajom@kol.km	Male	38	Medtronic Inc.	Health Therapist	(987) 619-2695	B6V 3W3	MB	5020000000000230	05/2018	
1	Miss	Claudia	Rodriquez	acu@jatsot.ug	Female	51	Ames Department Stores, Inc.	Health Therapist	(356) 736-7352	G7M 5F3	SK	5020000000000230	07/2028	
2	Miss	Harry	Meyer	zuz@lo.wf	Female	51	CSX Corp.	Political Scientist	(539) 246-1806	A0Z 6P9	NS	6300000000000000	02/2023	Sv
3	Miss	Edith	Gilbert	hansohsi@jupec.md	Female	55	Murphy Oil Corporation	Transportation Manager	(984) 962-7494	P9I 9H3	YT	3530000000000000	02/2028	Mac
4	Dr.	Lura	Murphy	webediti@je.st	Female	20	PETsMART Inc	Statistician	(902) 568-9748	S1A 6K0	ON	4030000000000000	10/2025	Diners Internati

◀

▶

In [3]:

Out[3]:

	prefix	first	last	email	gender	age	company	profession	phone	postal	province	cc_no	cc_exp	cc_
0	Dr.	Ray	Morton	sebvajom@kol.km	Male	38	Medtronic Inc.	Health Therapist	(987) 619-2695	B6V 3W3	MB	5020000000000230	05/2018	
1	Miss	Claudia	Rodriquez	acu@jatsot.ug	Female	51	Ames Department Stores, Inc.	Health Therapist	(356) 736-7352	G7M 5F3	SK	5020000000000230	07/2028	
2	Miss	Harry	Meyer	zuz@lo.wf	Female	51	CSX Corp.	Political Scientist	(539) 246-1806	A0Z 6P9	NS	6300000000000000	02/2023	Sv
3	Miss	Edith	Gilbert	hansohsi@jupec.md	Female	55	Murphy Oil Corporation	Transportation Manager	(984) 962-7494	P9I 9H3	YT	3530000000000000	02/2028	Mac
4	Dr.	Lura	Murphy	webediti@je.st	Female	20	PETsMART Inc	Statistician	(902) 568-9748	S1A 6K0	ON	4030000000000000	10/2025	Diners Internati

3. How many entries your data have? Can you tell the no. of columns in your data?

In [4]:

```
# Code here please
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 30000 entries, 0 to 29999
Data columns (total 20 columns):
 #   Column          Non-Null Count  Dtype
---  -
 0   prefix          30000 non-null  object
 1   first           30000 non-null  object
 2   last            30000 non-null  object
 3   email           30000 non-null  object
 4   gender          30000 non-null  object
 5   age             30000 non-null  int64
 6   company         30000 non-null  object
 7   profession      30000 non-null  object
 8   phone           30000 non-null  object
 9   postal          30000 non-null  object
10  province        30000 non-null  object
11  cc_no           30000 non-null  int64
12  cc_exp          30000 non-null  object
13  cc_type         30000 non-null  object
14  price(CAD)      30000 non-null  float64
15  fav_color       30000 non-null  object
16  ip              30000 non-null  object
17  weekday         30000 non-null  object
18  ampm            30000 non-null  object
19  date            30000 non-null  object
dtypes: float64(1), int64(2), object(17)
memory usage: 2.6+ MB
```

In [4]:

```
<class 'pandas.core.frame.DataFrame'>
,RangeIndex: 30000 entries, 0 to 29999
,Data columns (total 20 columns):
,prefix      30000 non-null object
,first       30000 non-null object
,last        30000 non-null object
,email       30000 non-null object
,gender      30000 non-null object
,age         30000 non-null int64
,company     30000 non-null object
,profession  30000 non-null object
,phone       30000 non-null object
,postal      30000 non-null object
,province    30000 non-null object
,cc_no       30000 non-null int64
,cc_exp      30000 non-null object
,cc_type     30000 non-null object
,price(CAD)  30000 non-null float64
,fav_color   30000 non-null object
,ip          30000 non-null object
,weekday     30000 non-null object
,ampm        30000 non-null object
,date        30000 non-null object
,dtypes: float64(1), int64(2), object(17)
,memory usage: 4.6+ MB
```

4. What are the max and min ages of your customer? Can you find mean of your customer?

```
In [5]: # Code here please
print("Max. age of the customer is:", df['age'].max())
print("Min. age of the customer is:", df['age'].min())
print("Avg. age of the customer is:", df['age'].mean())
```

```
Max. age of the customer is: 65
Min. age of the customer is: 18
Avg. age of the customer is: 41.550066666666666
```

```
In [5]:
Max. age of the customer is: 65
,Max. age of the customer is: 18
,Avg. age of the customer is: 41.550066666666666
```

5. What are the three most common customer's names?

✓ `value_counts()` returns object containing counts of unique values. The resulting object will be in descending order so that the first element is the most frequently-occurring element. Excludes NA values by default.

```
In [6]: # Code here please
df['first'].value_counts().head(3)
```

```
Out[6]: first
Willie    130
Francis   124
Eula      86
Name: count, dtype: int64
```

```
In [6]:
```

```
Out[6]: Willie    130
Francis   124
Eula      86
Name: first, dtype: int64
```

6. Two customers have the same phone number, can you find those customers?

```
In [8]: # Code here please
df['phone'].value_counts().head(2)
```

```
Out[8]: phone
(263) 382-8004    2
(712) 247-7037    1
Name: count, dtype: int64
```

```
In [7]: # Lets find what phone number is twice!
```

```
Out[7]: (263) 382-8004    2
(668) 836-2738    1
Name: phone, dtype: int64
```

```
In [9]: # Code here please
phone_counts = df['phone'].value_counts()
duplicate_mask = phone_counts > 1
duplicate_numbers = phone_counts[duplicate_mask].index
df[df['phone'].isin(duplicate_numbers)]
```

Out[9]:

	prefix	first	last	email	gender	age	company	profession	phone	postal	province	cc_no	cc_exp	cc_type	price(C
15	Mrs.	Lilly	Tyler	kofadu@itohi.tf	Female	38	CSX Corp.	Structural Engineer	(263) 382-8004	V7K 1E3	ON	3000000000000000	11/2022	Diners Club Carte Blanche	
16	Mrs.	Peter	Cain	megkosig@anazeor.gn	Male	27	Campbell Soup Co.	Insurance Adjuster	(263) 382-8004	E8T 2B4	YT	3530000000000000	03/2024	Solo	1

In [8]:

```
# Now we know the phone number, Let's find out the other stuff!
```

Out[8]:

	prefix	first	last	email	gender	age	company	profession	phone	postal	province	cc_no	cc_exp	cc_type	price(C
15	Mrs.	Lilly	Tyler	kofadu@itohi.tf	Female	38	CSX Corp.	Structural Engineer	(263) 382-8004	V7K 1E3	ON	3000000000000000	11/2022	Diners Club Carte Blanche	
16	Mrs.	Peter	Cain	megkosig@anazeor.gn	Male	27	Campbell Soup Co.	Insurance Adjuster	(263) 382-8004	E8T 2B4	YT	3530000000000000	03/2024	Solo	1

7. How many customers have profession "Structural Engineer"?

In [10]:

```
# Code here please
df[df['profession'] == 'Structural Engineer'].count()
```

```
Out[10]: prefix      87
          first      87
          last       87
          email      87
          gender     87
          age        87
          company    87
          profession 87
          phone      87
          postal     87
          province   87
          cc_no      87
          cc_exp     87
          cc_type    87
          price(CAD) 87
          fav_color  87
          ip         87
          weekday    87
          ampm       87
          date       87
          dtype: int64
```

In [9]:

```
Out[9]: prefix      87
          first      87
          last       87
          email      87
          gender     87
          age        87
          company    87
          profession 87
          phone      87
          postal     87
          province   87
          cc_no      87
          cc_exp     87
          cc_type    87
          price(CAD) 87
          fav_color  87
          ip         87
          weekday    87
          ampm       87
          date       87
          dtype: int64
```


8. How many male customers are 'Structural Engineer'?

```
In [11]: # Code here please
df[(df['profession'] == 'Structural Engineer') & (df['gender'] == 'Male')].count()
```

```
Out[11]: prefix      43
first      43
last       43
email      43
gender     43
age        43
company    43
profession 43
phone      43
postal     43
province   43
cc_no      43
cc_exp     43
cc_type    43
price(CAD) 43
fav_color  43
ip         43
weekday    43
ampm       43
date       43
dtype: int64
```

```
In [10]:
```

```
Out[10]: prefix      43
first      43
last       43
email      43
gender     43
age        43
company    43
profession 43
phone      43
postal     43
province   43
cc_no      43
cc_exp     43
cc_type    43
price(CAD) 43
fav_color  43
ip         43
weekday    43
ampm       43
date       43
dtype: int64
```

9. Find out the female Structural Engineers from province Alberta (AB)?

```
In [12]: # Code here please
```

```
df[(df['profession'] == 'Structural Engineer') & (df['gender'] == 'Female') & (df['province'] == 'AB')]
```

Out[12]:	prefix	first	last	email	gender	age	company	profession	phone	postal	province	cc_no	cc_exp	cc_ty
8858	Dr.	Roy	Stanley	apiunvo@ehasom.ir	Female	39	Cincinnati Financial Corp.	Structural Engineer	(876) 758-2929	N5A 6L2	AB	6300000000000000	02/2022	InstaPayme
9058	Mr.	Lora	Kennedy	dennap@rabac.se	Female	51	BJ Services Company	Structural Engineer	(305) 786-6959	N3R 4B7	AB	2010000000000000	06/2019	Diners C United Sta & Cana
24736	Mrs.	Nell	Richards	dob@me.ki	Female	18	Telephone & Data Systems Inc	Structural Engineer	(262) 681-5018	N3Y 3G2	AB	3470000000000000	07/2028	America Expr
29865	Mrs.	Don	McDaniel	naf@zudu.bj	Female	37	Anadarko Petroleum Corporation	Structural Engineer	(238) 789-2825	B4A 8Q9	AB	5610000000000000	12/2022	Mastercard



In [11]:															
Out[11]:	prefix	first	last	email	gender	age	company	profession	phone	postal	province		cc_no	cc_exp	cc_ty
	8858	Dr.	Roy	Stanley	apiunvo@ehasom.ir	Female	39	Cincinnati Financial Corp.	Structural Engineer	(876) 758-2929	N5A 6L2	AB	6300000000000000	02/2022	InstaPaym
	9058	Mr.	Lora	Kennedy	dennap@rabac.se	Female	51	BJ Services Company	Structural Engineer	(305) 786-6959	N3R 4B7	AB	2010000000000000	06/2019	Diners C United Sta & Cana
	24736	Mrs.	Nell	Richards	dob@me.ki	Female	18	Telephone & Data Systems Inc	Structural Engineer	(262) 681-5018	N3Y 3G2	AB	3470000000000000	07/2028	Americ Expr
	29865	Mrs.	Don	McDaniel	naf@zudu.bj	Female	37	Anadarko Petroleum Corporation	Structural Engineer	(238) 789-2825	B4A 8Q9	AB	5610000000000000	12/2022	Masterc

10. What is the max, min and average spending?

```
In [13]: # Code here please
print("Max. spending:", df['price(CAD)'].max())
print("Min. spending:", df['price(CAD)'].min())
print("Avg. spending:", df['price(CAD)'].mean())
```

Max. spending: 100.0
 Min. spending: 0.0
 Avg. spending: 49.990775

```
In [12]:
Max. spending: 100.0
,Mai. spending: 0.0
,Avg. spending: 49.990775000000184
```

11. Who did not spend anything? Company wants to send a deal to encourage the customer to buy stuff!

```
In [14]: # Code here please
df[df['price(CAD)'] == 0]
```

Out[14]:

	prefix	first	last	email	gender	age	company	profession	phone	postal	province	cc_no	cc_exp	cc_type	price(CA
5320	Dr.	Bruce	Bryan	ru@pubuspuh.cl	Male	59	Wal-Mart Stores Inc	Engineer Technician	(709) 446-8317	H6H 3Y0	NU	2010000000000000	04/2024	Bankcard	
10597	Mrs.	Flora	Clark	wad@me.com	Female	27	MetLife Inc.	Cruise Director	(775) 373-6590	B2U 8K6	NB	3010000000000000	05/2020	Diners Club Carte Blanche	

In [13]:

Out[13]:

	prefix	first	last	email	gender	age	company	profession	phone	postal	province	cc_no	cc_exp	cc_type	price(CA
5320	Dr.	Bruce	Bryan	ru@pubuspuh.cl	Male	59	Wal-Mart Stores Inc	Engineer Technician	(709) 446-8317	H6H 3Y0	NU	2010000000000000	04/2024	Bankcard	
10597	Mrs.	Flora	Clark	wad@me.com	Female	27	MetLife Inc.	Cruise Director	(775) 373-6590	B2U 8K6	NB	3010000000000000	05/2020	Diners Club Carte Blanche	

12. As a loyalty reward, company wants to send thanks coupon to those who spent 100CAD or more, please find out the customers?

In [15]:

```
# Code here please
df[df['price(CAD)'] >= 100]
```

Out[15]:

	prefix	first	last	email	gender	age	company	profession	phone	postal	province	cc_no	cc_exp	cc_t
76	Mrs.	Gregory	Brown	hav@jek.gs	Female	31	E*Trade Group, Inc.	Novelist	(625) 537-8923	X3S 4Q2	PE	6010000000000000	04/2018	
21093	Mrs.	Cody	Christensen	get@jovu.ag	Male	28	National City Corp.	Hospital Administrator	(261) 737-3292	X0S 1O5	PE	2010000000000000	03/2024	
24385	Miss	Lizzie	Dixon	goh@tuwjaz.gd	Female	38	FleetBoston Financial Co.	Compensation Analyst	(989) 239-1752	V8X 9V6	NB	6300000000000000	07/2019	Bankc

In [14]:

Out[14]:

	prefix	first	last	email	gender	age	company	profession	phone	postal	province	cc_no	cc_exp	cc_t
76	Mrs.	Gregory	Brown	hav@jek.gs	Female	31	E*Trade Group, Inc.	Novelist	(625) 537-8923	X3S 4Q2	PE	6010000000000000	04/2018	
21093	Mrs.	Cody	Christensen	get@jovu.ag	Male	28	National City Corp.	Hospital Administrator	(261) 737-3292	X0S 1O5	PE	2010000000000000	03/2024	
24385	Miss	Lizzie	Dixon	goh@tuwjaz.gd	Female	38	FleetBoston Financial Co.	Compensation Analyst	(989) 239-1752	V8X 9V6	NB	6300000000000000	07/2019	Bankc

13. How many emails are associated with this credit card number '502000000000230'?

In [16]:

```
# Code here please
df[df['cc_no'] == 502000000000230]['email']
```

Out[16]:

```
0    sebvajom@kol.km
1    acu@jatsot.ug
Name: email, dtype: object
```

In [15]:

```
Out[15]: 0    sebvajom@kol.km
          1    acu@jatsot.ug
          Name: email, dtype: object
```

14. We need to send new cards to the customers well before the expire, how many cards are expiring in 2019?

Use `sum()` and `count()` and see the difference in their use :)

```
In [18]: # Code here please
```

```
#--using count()--
# This counts the actual rows in the DataFrame that match the condition
#print(df[df['cc_exp'].str[-4:] == '2019']['cc_exp'].count())

#--using sum()--
# This creates a Boolean series (True/False). sum() treats True as 1 and False as 0
# So summing the series gives the total number of cards expiring in 2019
print((df['cc_exp'].str[-4:] == '2019').sum())
```

```
2684
```

```
In [16]:
```

```
Out[16]: 2684
```

15. How many people use Visa as their Credit Card Provider?

```
In [19]: # Code here please
print(df[df['cc_type'] == 'Visa']['cc_type'].count())
```

```
1721
```

```
In [17]:
```

```
Out[17]: 1721
```

16. Can you find the customer who spent 100 CAD using Visa?

```
In [20]: # Code here please
df[(df['cc_type'] == 'Visa') & (df['price(CAD)'] == 100)]
```

Out[20]:	prefix	first	last	email	gender	age	company	profession	phone	postal	province	cc_no	cc_exp	cc_type	price(CAD)	1
76	Mrs.	Gregory	Brown	hav@jek.gs	Female	31	E*Trade Group, Inc.	Novelist	(625) 537-8923	X3S 4Q2	PE	6010000000000000	04/2018	Visa	100.0	

In [18]:

Out[18]:	prefix	first	last	email	gender	age	company	profession	phone	postal	province	cc_no	cc_exp	cc_type	price(CAD)	1
76	Mrs.	Gregory	Brown	hav@jek.gs	Female	31	E*Trade Group, Inc.	Novelist	(625) 537-8923	X3S 4Q2	PE	6010000000000000	04/2018	Visa	100.0	

17. What are two most common professions?

```
In [21]: # Code here please
df['profession'].value_counts().head(2)
```

```
Out[21]: profession
Preschool Teacher    112
Distribution Manager  107
Name: count, dtype: int64
```

In [19]:

```
Out[19]: Preschool Teacher    112
Distribution Manager  107
Name: profession, dtype: int64
```

18. Can you tell the top 5 most popular email providers? (e.g. gmail.com, yahoo.com, etc...)

```
In [22]: # Code here please
df['email'].str.split('@').str[1].value_counts().head()
```

Out[22]: email
gmail.com 1687
me.com 1676
outlook.com 1664
live.com 1660
hotmail.com 1659
Name: count, dtype: int64

In [20]:

Out[20]: gmail.com 1687
me.com 1676
outlook.com 1664
live.com 1660
hotmail.com 1659
Name: email, dtype: int64

19. Is there any customer who is using email with "am.edu"?

Hint: Use `lambda` expression in `apply()` . split the email address at `@` .

In [23]: *# Code here please*
`df[df['email'].apply(lambda x: x.split('@')[1]) == 'am.edu']`

Out[23]:

	prefix	first	last	email	gender	age	company	profession	phone	postal	province	cc_no	cc_exp	cc_type	pric
150	Miss	Loretta	Fletcher	barit@am.edu	Female	48	International Corp	York Rehabilitation Counselor	(323) 279-8038	E5X 8L0	QC	6330000000000000	05/2021	Diners Club United States & Canada	

In [21]:

Out[21]:	prefix	first	last	email	gender	age	company	profession	phone	postal	province	cc_no	cc_exp	cc_type	pric
150	Miss	Loretta	Fletcher	barit@am.edu	Female	48	International Corp	Rehabilitation Counselor	(323) 279-8038	E5X 8L0	QC	6330000000000000	05/2021	Diners Club United States & Canada	

20. Which day of the week, the store gets more customers?

```
In [24]: # Code here please
df['weekday'].value_counts().head()
```

```
Out[24]: weekday
Saturday    4376
Wednesday   4365
Thursday    4327
Friday       4316
Monday       4216
Name: count, dtype: int64
```

```
In [22]:
```

```
Out[22]: Saturday    4376
Wednesday   4365
Thursday    4327
Friday       4316
Monday       4216
Name: weekday, dtype: int64
```

Excellent work!